

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circle Theorems

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CIRCLE THEOREMS · CH4 · L4

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The Map

This lesson collapses Circle Theorems into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Spot the theorem family

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use tangent-radius or alternate segment

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use cyclic quadrilateral angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Write the reason with the answer

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use circle theorem

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 31 website questions, 31 checked solutions, 5 local PDFs read.

01 Spot the theorem family

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q14

Angle and central angle stand

This is a reasoning step: write one clear sentence, then continue the calculation.

The angle at the centre

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Finish: 52° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

02 Use tangent-radius or alternate segment

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle BFD = 39^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q14

Use trigonometry

$$\angle BFD = 39^\circ$$

Given

$$\angle EFD = 18^\circ$$

Finish: 123° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

03 Use cyclic quadrilateral angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2021 P2H Q18

Use trigonometry

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

Opposite angles in a cyclic

$$\angle ADC = 180^\circ - 98^\circ = 82^\circ$$

Finish: 32.8 cm.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

04 Write the reason with the answer

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle RQP = 90^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q14

Use trigonometry

$$\angle RQP = 90^\circ$$

Reason the angle in a

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\angle RQP = 90^\circ$, and $\angle PSQ = 22^\circ$.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q14

Angle and central angle stand

This is a reasoning step: write one clear sentence, then continue the calculation.

The angle at the centre

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Finish: 52° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

06 Use circle theorem

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m = \frac{124}{2} = 62$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q13

Use trigonometry

$$m = \frac{124}{2} = 62$$

Use circle theorem

$$n = 180 - 62 = 118$$

Finish: $m = 62$, $n = 118$, and $\angle QPO = 62^\circ$.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Spot the theorem family
"work out", "show", "hence", a familiar exam phrase	Use tangent-radius or alternate segment
"work out", "show", "hence", a familiar exam phrase	Use cyclic quadrilateral angles
"work out", "show", "hence", a familiar exam phrase	Write the reason with the answer
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use circle theorem
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.